Introduction

An added-variable plot (AVP), also called a partial regression plot, displays the relationship between a single predictor and the outcome after adjusting both for all other predictors in the model. It is the geometric expression of a regression coefficient: the slope of the OLS line in the AVP equals the coefficient in the full multiple regression. AVPs are the standard tool for diagnosing whether a partial linear effect is well-supported, whether non-linearity has been missed, and whether one or two observations are driving a particular coefficient.

Prerequisites

A working understanding of multiple regression, partial correlation, and the residualisation interpretation of partial effects.

Theory

For predictor X_j in a model with other predictors X_{-j}:

1. Regress Y on X_{-j} and save residuals $e_{Y|X_{-j}}$.
2. Regress X_j on X_{-j} and save residuals $e_{X_j|X_{-j}}$.
3. Plot $e_{Y|X_{-j}}$ against $e_{X_j|X_{-j}}$.

The OLS slope of this plot equals the coefficient on X_j in the full multiple regression — the famous Frisch-Waugh-Lovell theorem. Patterns in the AVP reveal partial-effect non-linearity, leverage, and outliers specific to X_j that bivariate scatter plots cannot show.

Assumptions

Standard multiple regression assumptions; the residualisation interpretation requires that the other predictors are themselves correctly specified.

R Implementation

```r
library(car)

fit <- lm(mpg ~ wt + hp + disp + drat, data = mtcars)
avPlots(fit)

# Single AVP
avPlot(fit, variable = "hp")
```

Output & Results

`avPlots()` draws one scatter plot per predictor, each with the fitted partial regression line. The slope of each line equals the corresponding multiple-regression coefficient. Outlier and leverage diagnostics specific to each predictor become visible.

Interpretation

A reporting sentence: "The added-variable plot for horsepower showed a clear negative slope after adjusting for weight, displacement, and rear-axle ratio, with no evidence of non-linearity or individual influential observations; the AVP slope of -0.030 matches the multiple-regression coefficient ($p = 0.005$). Diagnostic plots support reporting the linear partial effect of horsepower." Use AVPs to check both the shape of a partial relationship and to flag observations driving specific coefficients.

Practical Tips

- AVPs are more informative than raw bivariate scatter plots for checking partial effects; bivariate plots ignore the other predictors and can mislead.
- Non-linear patterns in an AVP suggest a transformation, polynomial term, or spline for that specific predictor.
- Outliers in an AVP indicate observations influencing that particular coefficient; cross-check with DFBETAs.
- `car::avPlots(fit)` draws all AVPs in a single grid for compact diagnostic display.
- Distinguish from component-plus-residual plots (`crPlots`), which add the predictor's full effect rather than partialling out other predictors; the two answer related but different diagnostic questions.
- For GLMs and mixed models, AVP analogues exist; see `car::avPlots()` and `effects::predictorEffects()`.

R Packages Used

`car::avPlots()` for the canonical implementation; `car::crPlots()` for component-plus-residual plots; `effects::predictorEffects()` for marginal-effect plots; `ggeffects::ggpredict()` for ggplot-style predicted-effect plots that complement AVP diagnostics.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Correlation vs regression; Model-I/II regression
- Simple linear regression
- Multiple regression: confounding, interaction, centring
- Diagnostics: residuals, QQ, leverage, Cook's distance, VIF
- Robust regression, weighted LS, HC (sandwich) standard errors
- One-way ANOVA and contrasts (emmeans)
- Two-way / factorial ANOVA with interaction
- RCBD and repeated measures → mixed models
- GAMs with mgcv
- Non-linear regression with nls
- Logistic regression (binomial GLM)
- ANCOVA in RCTs (baseline adjustment)
- Ordinal and multinomial regression
- Poisson and negative-binomial regression
- Calibration, discrimination, ROC/AUC, Brier score
- Dichotomisation, change scores, regression to the mean
- Decision curves, NRI, IDI